The documentation and process conversion measures necessary to comply with this amendment shall be completed by 20 Feb 93

INCH-POUND

MIL-S-19500/297B
AMENDMENT 2
20 October 1992
SUPERSEDING
AMENDMENT 1
3 July 1990

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER,
TYPES 1N1184, 1N1186, 1N1188, 1N1190, 1N3766, 1N3768,
1N1184R, 1N1186R, 1N1188R, 1N1190R, 1N3766R, 1N3768R,
JAN, JANTX, AND JANTXV

This amendment forms a part of MIL-S-19500/297B, dated 30 March 1989, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 4

4.3, screening table, screen number 11: Delete "ΔV_{F2} = 1 V(pk)" and substitute "ΔV_{F2} = 0.1 V(pk)". Delete "ΔI_{R1} = ±40 μA dc" and substitute "ΔI_{R1} = ±5 μA dc".

* 4.3.2: Delete and substitute as follows:

"4.3.2 Thermal response. The thermal response measurements shall be performed in accordance with method 3101 of MIL-STD-750. The thermal response conditions and maximum thermal response limit shall be derived by each vendor. The chosen thermal response measurement and conditions for each device in the qualification lot shall be submitted in the qualification report and a thermal response curve shall be plotted. The chosen thermal response values shall be considered final after the manufacturer has had the opportunity to test five consecutive lots. Heating current (I_M) ≥ rated I_O; t_H = 150 to 400 ms; t_{MD} = 50 to 300 μs; 50 mA ≤ I_M ≤ 250 mA."

PAGE 6

* TABLE I, Subgroup 2: Delete the following:

"Forward voltage	4011	$I_F = i_{f(surge)}$ t_p = 800 μs		V_{F1}	2.5	V dc

TABLE I: Add the following.

Subgroup 6						
Forward voltage 2/	4011	$I_F = i_{f(surge)}$ t_p = 800 μs		V_{F1}	2.5	V dc
Forward voltage 2/	4011	$I_F = i_{f(surge)}$ t_p = 8.3 ms		V_{F1}	2.3	V dc"

2/ V_{F1} shall be performed with either t_p = 800 μs or t_p 8.3 ms.

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PAGE 10

TABLE IV, subgroup 3, DPA, method column: Add "2101".

* TABLE IV, subgroup 4, thermal resistance, conditions column, delete and substitute:

"$R_{\theta JC}$ = 0.8°C/W maximum
I_M = 50 mA to 250 mA
I_H ≥ rated I_O
t_H = 150 to 400 ms
t_{MD} = 50 to 300 µs"

PAGE 11

TABLE V, delete table title and substitute: "TABLE V. Groups A, B, C, and E electrical end-point measurements."

TABLE V, delete footnote.

NOTE: The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Air Force - 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC, OS
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5961-1359)

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